

In the previous class, you have studied that sound is a form of energy that is produced by the vibrating objects. Sound needs a medium to travel from one place to another. The properties of a medium determine the audibility and quality of sound. Did you ever experience the change in the voice of your friends on the phone as compared to their voices in person?

As we know, water ripples can be produced by throwing a stone in water (as shown in figure 10.1). What cause waves? It is the vibration of water molecules that causes wave. We can define these ripples or waves as the disturbance produced by the vibration of different substances.

In this Chapter you will learn about:

- Transverse and longitudinal waves.
- Introduction to terms (wavelength, speed, amplitude and frequency)
- Pitch and loudness
- Audible frequency range
- Applications of different sounds in our daily life (doorbells, sirens, telephones, radios, stereos, smoke detectors security system alarms)

All the students will be able to:

- ✓ Explain the wavelength, frequency and amplitude of sound and give their units.
- ✓ State factors on which sound depends.
- ✓ Investigate objects in home and surroundings that are designed and made to produce different sounds.
- ✓ Compare audible frequency range of human beings and different animals.
- ✓ Design a musical instrument to explain the relation between its sound and shape.
- ✓ Identify the applications of different sounds in daily life.



Figure 10.1: Water ripples

TRANSVERSE AND LONGITUDINAL WAVES

- ✓ Explain the wavelength, frequency and amplitude of sound and give their units.

Waves transfer energy from one place to another. However, different forms of energy travel through different waves. These waves have distinctive characteristics. For instance, sound waves need a medium (some material such as metals, water and air) to travel.

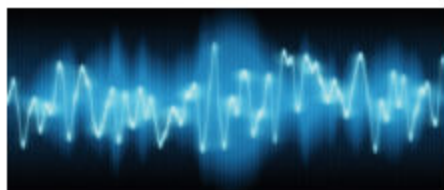


Figure 10.2: Sound waves

Activity 10.1: Investigate the properties of different waves.



What I need:

Two slinkies (plastic coil/spring), pencil/pen.

What to do:

Stand in front of each other and hold the slinky as shown in figure 10.3 (a).

Pair A: First, compress the slinky with full force.

Note: It will compress to the level, it sticks to other member's hand as shown in figure 10.3 (b).

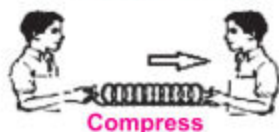
Now, release the pressure on the slinky.

Pair B: Move slinky up and down with a greater speed. **Figure 10.3 (b)**

Observe the movement of slinky produced by pair A and B respectively. Take notes to answer the questions, mentioned in the observation column.



Figure 10.3 (a): Students holding a slinky



What I observed:

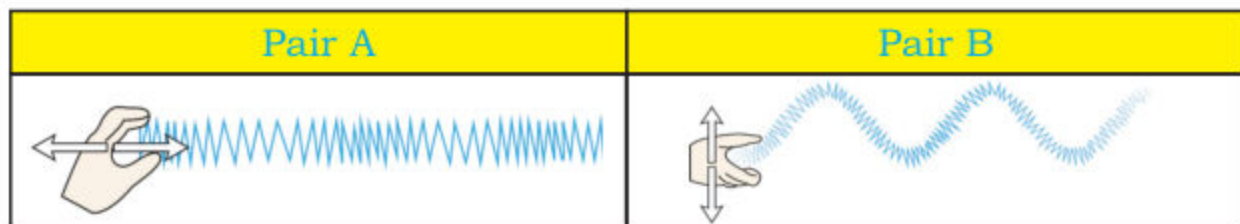


Figure 10.3 (c): Direction of hand movement and waves formation

- Is the vibration of slinky perpendicular or parallel to the wave propagation?
- Are there any areas of compression in the slinky?
- How many waves are produced during one movement of the slinky?

Activity Questions:

1. How many types of waves can be produced through vertical and horizontal movements of the slinky?
2. What are the characteristics of the waves shown in figure 10.3 (c)?

Waves produced by pair 'A' have areas of compression and rarefaction, and the movement of the particles in slinky seems parallel to the direction of wave propagation. In contrast, waves produced by pair 'B' have crests and troughs and the movement of particles in slinky seems perpendicular to the direction of wave propagation.

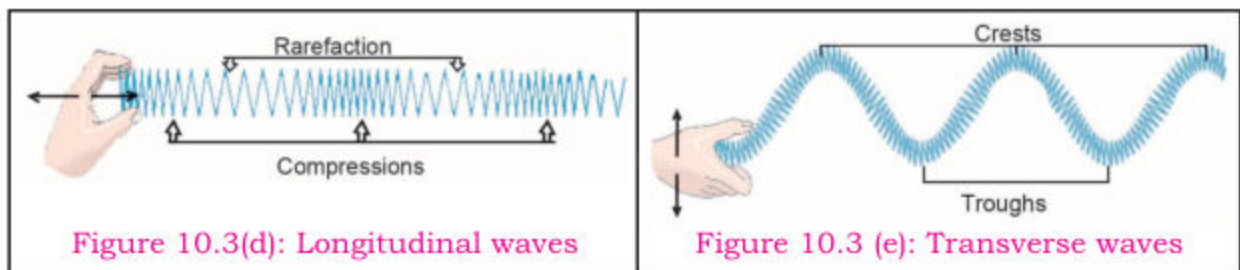


Figure 10.3(d): Longitudinal waves

Figure 10.3 (e): Transverse waves

Waves having compressions and rarefactions (as shown in figure 10.3 d) with oscillation (movements) of particles parallel to the direction of wave propagation are named as **longitudinal waves**. Examples are seismic and sound waves. In contrast, waves having crests and troughs (as shown in figure 10.3 e) with oscillation (movements) of particles perpendicular to the direction of wave

propagation are called **transverse waves**.

In solids, sound waves are transverse waves because molecules just vibrate on their rest position and pass energy from one point to another. However, intermolecular spaces in fluids and gases allow only longitudinal sound waves to travel through them. Furthermore, the speed and audibility of sound also depend on the molecular movements of its medium.

In order to explain the characteristics of sound waves, we need to learn different terms such as:

- (i) wavelength,
- (ii) frequency,
- (iii) amplitude and,
- (iv) speed of these waves.

(i) **Wavelength:**

In a transverse wave, the wavelength is the distance between its two adjacent crests or troughs as shown in figure 10.4. Similarly, the wavelength is the distance between two adjacent compressions or rarefactions of a longitudinal wave. Wavelength is symbolized by the Greek letter, lambda (λ) while it is measured in metres (m).

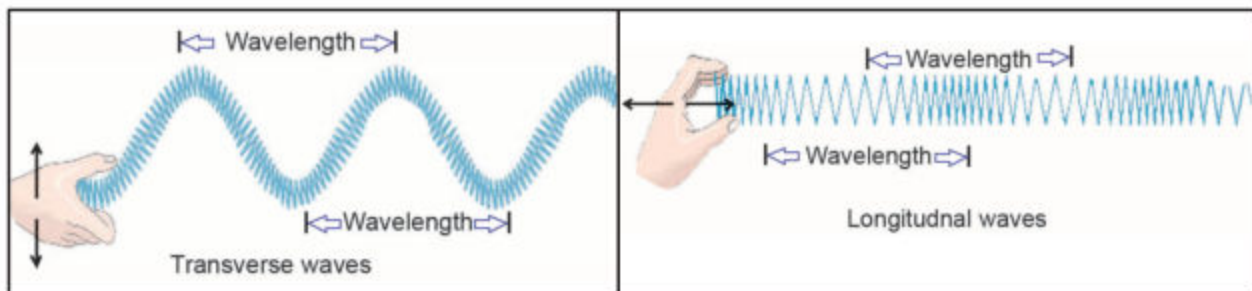


Figure 10.4: Sound waves

(ii) **Amplitude:**

As waves travel, the distance between the rest position of the moving particles to the top of a crest or the bottom of a trough is termed as amplitude (a). Amplitude is measured in metre (m) as well as decibel (dB) of the sound pressure. Here it is important to understand that amplitude is not the distance between the crest (top) and the trough (bottom) of a wave. The difference between amplitude (a) and wavelength (λ) is evident in figure 10.5.

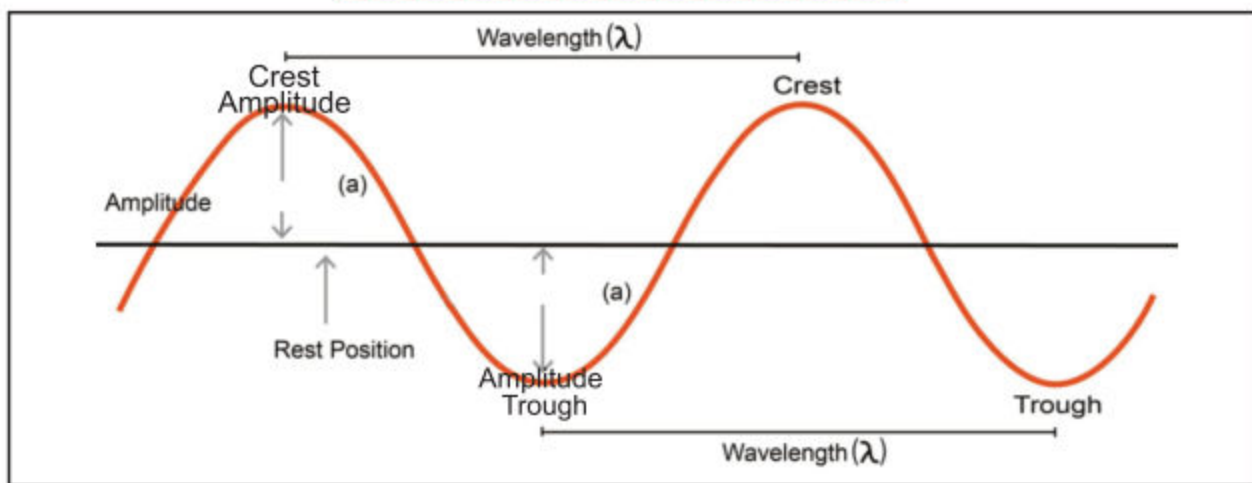


Figure 10.5: Amplitude (a) and Wavelength (λ)

(iii) Frequency:

The number of vibrations produced by a vibrating body in a second is called frequency (**f**). Frequency is measured in Hertz (Hz). Frequency of the sound waves can be calculated by the following formula:

$$\text{Frequency (f)} = \frac{\text{Number of waves}}{\text{Time (in seconds)}}$$

For instance, when one wave passes through a point in one second, the frequency will be 1Hz (figure 10. 6). It will be mentioned as 1cycle/s. Here, 's' denotes time in seconds.

(iv) Speed:

The distance a sound wave covers in a unit of time is called its speed. Speed is measured in metre per second (m/s). Both light and sound are forms of energy that have different speed. Light waves travel faster as compared to the sound waves. The speed of sound also depends on the properties of its medium, hence sound travels at a different speed in different mediums. For instance, sound travels fastest in diamond and slowest in air. The stiffer the medium, the faster is the speed of sound.

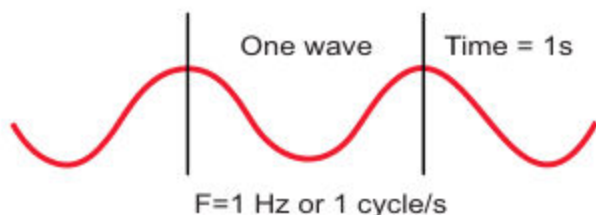


Figure 10.6: Frequency of a sound wave



Figure 10.6 Lightning and thunder

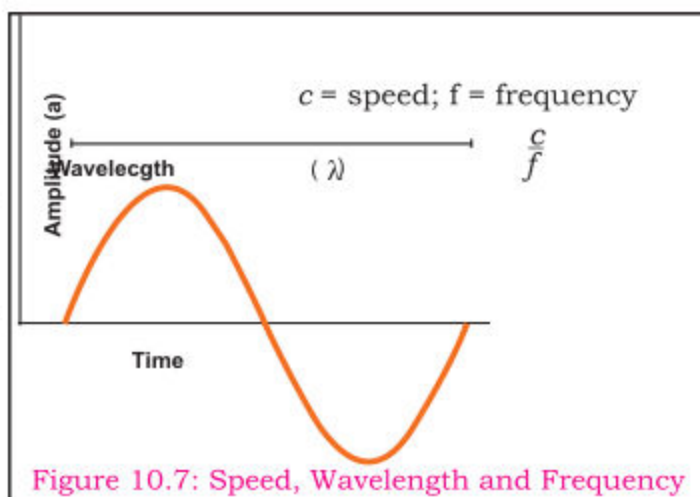
Table 10.1 shows the speed of sound in different media.

Medium	Speed	Medium	Speed
Air	343 m/s	Diamond	12000 m/s
Water	1433 m/s	Brick	4176 m/s
Steel	6100 m/s	Wood	3300-3600 m/s

RELATIONSHIP BETWEEN SPEED, WAVELENGTH AND FREQUENCY OF SOUND WAVES

✓ State factors on which sound depends

In our daily life, we come across with different types of sounds ranging from traffic noise to the melodious music. Can you find the reasons behind such a wide range of sounds we hear in our daily routine? The speed of sound is mathematically related to the wavelength and the frequency of its waves (as shown in figure 10.7).



The quality of sound depends on the properties of its medium, frequency, amplitude and speed of its waves. These factors have an impact on the audibility and pleasantness of sound. Interestingly, whether it is the unpleasant and irritating noise of traffic or the melody of our national anthem, pitch and loudness are the characteristics on which these sounds depend.

PITCH AND LOUDNESS:

PITCH:

Pitch is the highness and lowness of sound that is determined by the rate of the vibrations or frequency of sound waves. As pitch is the perceptual property of sound, high frequency of sound waves results into high-pitched, (thin sound) while a low frequency of sound waves results in low pitched, (heavy sound). Females and children have a thin and delicate voice as compared to the adult males.

Activity 10.2: Explore the effect of speed, frequency and pitch of sound.

What I need:

One set of (a rectangular plastic/metallic pencil box and three rubber bands of the same length but of three visibly distinguished thicknesses), pencil/pen.

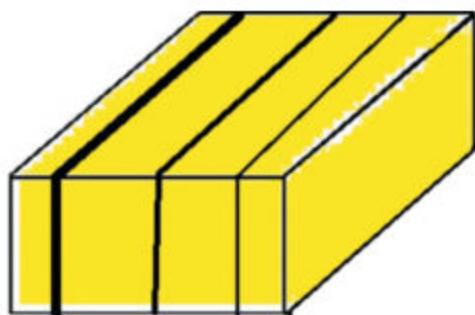


Figure 10.8 (a): Box for Activity 10.

What to do:

Stretch each rubber band around the box as shown in figure 10.8 (a).

Mark the thinnest rubber band as 'A', middle as 'B' and the thickest as 'C'.

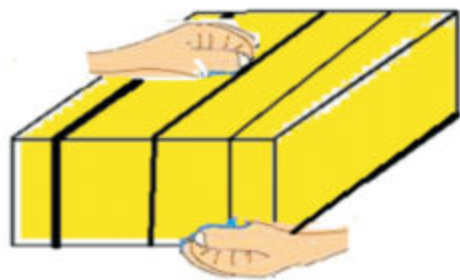


Figure 10.8 (b): How to hold the box and fret the rubber band

At first, hold the box in one hand and fret the index finger of the other hand on these rubber bands as one plays on the strings of a guitar[as shown in figure 10.8 (b)]. Increase the speed of fretting and note the variations in the pitch of the sound.

Pull each rubber band and note the variations in pitch (highness and lowness of sound).

Now, hold rubber band 'B' tightly in the middle and pull the rubber band on either side of the position, it is held. Note the pitch of the sound, produced in this act.

Change the position of your hold on the rubber to increase and decrease the length of the vibrating rubber band. Note the variations in the pitch for the difference in the length of the vibrating rubber band.

What I observed:

Activity Questions:

What does happen to the pitch of the sound as the thickness of rubber band increases?	Rubber band 'A'	Rubber band 'B'	Rubber band 'C'
What does happen to the pitch of the sound as the length of vibrating rubber band increases?	Positions of the Vibrating rubber band 'B'		
	Middle	Increased	Decreased
What happens to the pitch of sound when the speed of fretting fingers increases?			

LOUDNESS

- ✓ Investigate objects in home and surroundings that are designed and made to produce different sounds.

Loudness is “the degree of the sensation of sound produced in the human ear”. Loudness basically depends on the amplitude of the sound waves. However, the surface area of the source of sound and its distance from the listener also impact the loudness of the sound.

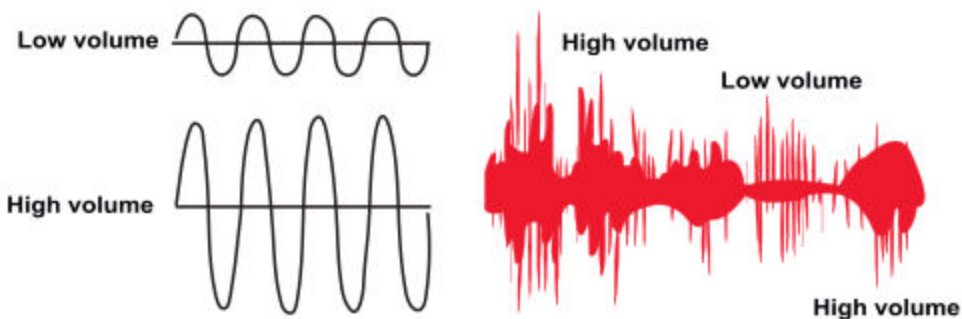


Figure 10.9: Loudness of Sound: (a) Graphics; (b) Cluster

Teacher Note: Divide students into groups and provide one set of the materials to each group. One member of each group will pull each rubber to different lengths, while other members of the group will record their observations in the above-stated columns. Make sure that rubber band should be tightly wrapped around the box.

In addition, one needs a greater amount of energy to produce louder and huskier sound. In other words, a greater amount of energy results into the higher amplitude of the sound waves which in turn increases the loudness and the intensity of the sound.

Activity 10.3: Investigate objects in home and surroundings that are designed and made to produce different sounds.

What I need:

Pencil/pen

What to do:

- Think of different sounds you heard since morning.
- Mention the source of the sound and record the observations regarding the difference in their pitch and loudness in table 10.2.

Note: Select only six different types of sounds.

Activity Questions	What I observe:		
Name the source of sound that produce different sounds.	Mention the quality of sound and its pitch and loudness.		
	Thinness or heaviness of sound	Pitch of sound	Loudness of sound
School bell	Heavy	Low	Loud
If the sound is heavy, what can be the contributory factors for it?			
Answer:			
If the sound is thin, what can be the contributory factors for it?			
Answer:			
Is there any relationship between the quality of sound pitch and loudness?			
Answer:			

Table 10.2: Exploring the difference in the characteristics on which sound depends

AUDIBLE FREQUENCY RANGE

✓ Compare audible frequency range of human beings and different animals

Pitch and loudness determine the quality and intensity of sound. Here, the quality and intensity of sound denote heavy or thin, loud or slow, and clearly audible sounds of different animals and humans. Can you define audible sound?

Audible sound refers to the sound that can be heard. In this way, the audible sound has a range of frequencies which can be heard by an animal or human. However, audible frequency range varies for humans and different species of animals.

Table 10.3 shows the audible frequency ranges of humans and the different species of animals.

Animal	Audible frequency range	Animals	Audible frequency range
Human	20-20,000 Hz	Cat	45-64,000 Hz
Elephant	16-20,000 Hz	Dolphin	20-120,000 Hz
Whale	10-31,000 Hz	Bat	2,000-110,000 Hz
Dog	67-45,000 Hz		

Activity 10.4: Design a musical instrument to explain the relation between its sound and shape.

What I need:

Set of (two cardboards, blue/black marker, scissor, a 12-inch ruler, scotch tape and gum stick).

What to do:

- Use a ruler to measure the length of the two cardboards.
- Use saw to cut 1 inch from one end of each cardboard and save cardboard pieces to make the stoppers.
- Stopper for round flute: Use one cardboard piece to draw a circle.
- Cut around inside of the circle.
- Stopper for rectangular flute: Use another cardboard piece to draw a rectangle. Cut inside of the rectangle.
- Measure 2 inches from one end of each cardboard and mark it.

Continue the line another 0.7 inch and mark again as shown in figure 10.10 (a). This is the area where wind-way will be cut down.

Continue the line another 3 inches and mark, then 7 millimetre (mm) and mark again. This 7 mm is the first hole where the first finger will be placed during playing the flute. From this point measure 0.7 inches between each 7mm area for next five holes as shown in figure 10.10(b)

Fold one cardboard into a circular way while the other into a rectangular way.

Use scotch tape to secure the circular and rectangular shapes of the flutes as shown in figure 10.10 (b).

Place round stopper inside the end of the round tube. Then, put some glue around the inner edges of the stopper and press the edges to get them stuck. Leave this tube to dry thoroughly as shown in figure 10.10 (c).

Repeat the process with the rectangular tube.

Play flute one by one and note the difference between the sounds produced by these two flutes in table 10.4.



Figure 10.10 (a): How to make wind-way



Figure 10.10 (b): How to make Marks

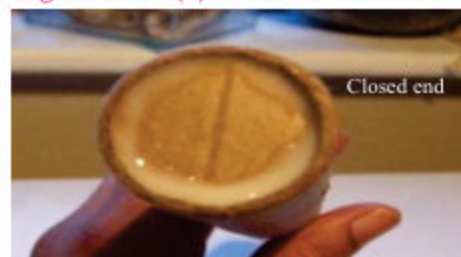


Figure 10.10 (c): How to close one end of the round tube (flute)

What I observe:

Table 10.4: Exploring the relation between the shape of the musical instrument and its sound.

Source of sound	difference in the quality of sound
Flute (round tube)	
Flute (rectangular tube)	

Activity Questions:

1. How does the shape of the flute affect the quality of its sound?
2. Enlist the differences in the quality of sound, produced by the two flutes.
3. Which flute makes the highest pitch?
4. Which flute makes the lowest pitch?

APPLICATIONS OF DIFFERENT SOUNDS

✓ Identify the applications of different sounds in daily life.

Sound as a form of energy has multiple applications in our daily life. Ranging from the ultra sonic sound to the infrasound, all frequencies of sound have distinctive benefits for humans. Sounds are used for the safety and security; treatment and detection of different diseases; telecommunications and entertainment of humans in their daily life. For example horn, Azan, bell, traffic, announcement and findings.

Sound waves in routine practices:



Figure 10.11 (a): Doorbell

The sound of doorbell indicates the presence of someone on the other side of the door.



Figure 10.11 (b): Home security alarm

The sound of home security alarm attracts the attention of its residents to the danger.



Figure 10.11 (c): Smoke detector

The alarming beep of smoke detector attracts our attention to the source of the smoke for safety and security reasons.



Figure 10.11 (d): Mobile phone

The sound of mobile phone attracts our attention to the distant call.



Figure 10.11 (e): Siren

Siren is used at multiple occasions to attract people for different reasons. For instance, it is used to warn us about incoming danger.



Figure 10.11 (f): Parking sensor system

Parking sensor system uses ultrasonic detectors that produce loud sound waves on the collision of any object with the car. This system warns the driver about the distance and direction of the object that hits her/his car.



Figure 10.11 (g): Music system

Music system entertains us through playing music at adjustable volumes.



Figure 10.11 (h): Radio

Radio is used for the transmission of sounds (voice and music).

Summary



Sound waves travel through solids, liquids and gases at different speed.

Compression

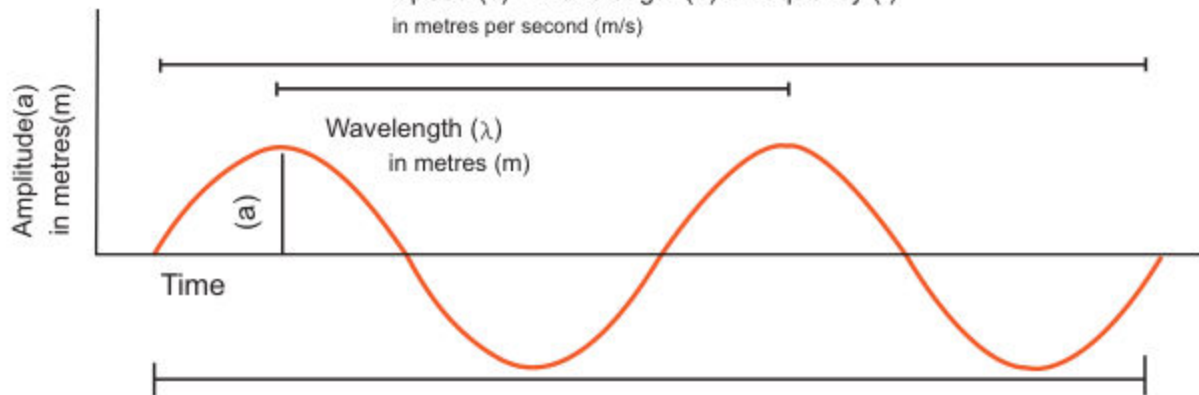
Rarefractions

Sound waves are **transverse waves** in solids.

Sound waves are **longitudinal waves** in liquids and gases.

Terms to explain Sound Waves and their Attributes

Speed (c) = wavelength (λ) $\times$ frequency (f)
in metres per second (m/s)



Frequency (f) = $\frac{\text{Number of waves}}{\text{Times (in seconds)}}$
in hertz (Hz)

Pitch and loudness are the characteristics of sound

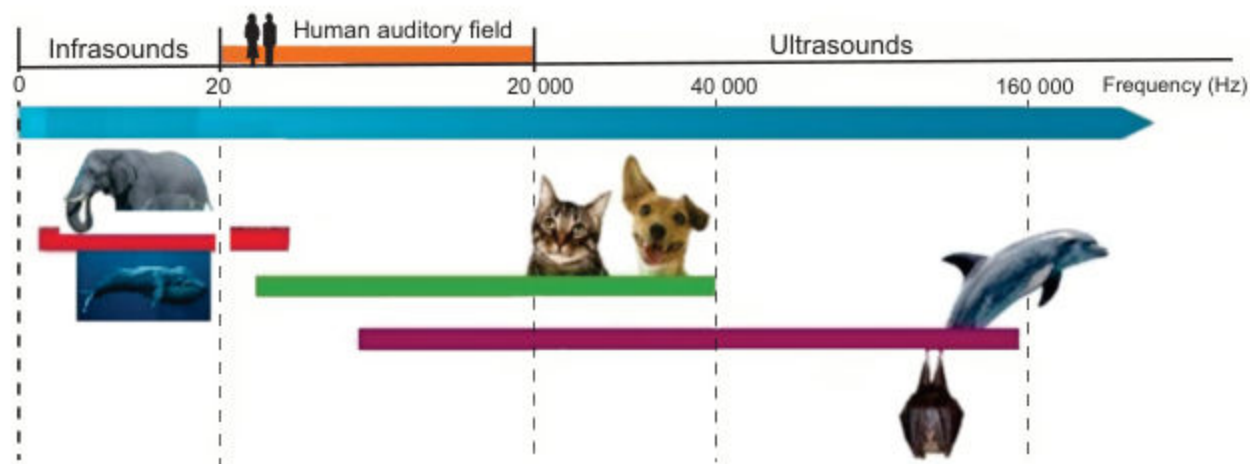
Pitch is the highness and lowness of sound.

Loudness is the degree of the sensation of sound produced in the human ear.

Applications of Sound



Audible Frequency ranges of Human and Animals Sound:



Review Exercises

1. Circle the most appropriate option:

a) Sound waves travel in solvents through _____.

- (i) transverse waves (ii) longitudinal waves
(iii) Both the transverse and longitudinal waves

b) Wavelength is symbolised as _____.

- (i) v (ii) λ (iii) δ

c) A normal human ear can hear sounds of the frequency ranges _____.

- (i) 6-20,000 Hz. (ii) 20-20,000 Hz. (iii) 20-3,000 Hz.

d) Sound travels fastest in the _____.

- (i) air (ii) water (iii) wood

e) Decibel is the unit of measurement of _____.

- (i) pitch (ii) amplitude (iii) wavelength

2. Explain the factors that impact the quality and intensity of sound.

3. Describe five ways in which sound is important for humans.

4. Given to the sound frequency ranges, identify the audible frequency ranges to a young woman, an old man, bats, cats, whale, dogs, dolphins and elephants:

Sound Frequency Ranges	Audible Frequency Ranges to
20-15,000 Hz	
2,000-110,000 Hz	
20-120,000 Hz	
45-64000 Hz	

PROJECT

Knowledge

Quality, intensity and audibility of sound depend on the wavelength, amplitude, frequency and speed of its waves. In addition, a musical instrument's shape also affects the quality and intensity of its music (sound). Like the flute, lab-based xylophone also demonstrates the relation between the quality and intensity of sound.

Let us make a simple xylophone to demonstrate the relation between the quality of sound and the shape of the source of sound!

Apparatus and Materials required:

- Five 1-litre water bottles/plastic jars with flat bottom and wide mouth
- Cutter
- One inch wide scotch tape
- Metal spoon
- 3 litres water
- Measuring cylinder

Procedure:

1. Cut bottles/jars into five sizes in a way that two adjacent bottles must have a 1-inch difference in their heights.
2. Mark these bottles/jars as I, II, III, IV and V.
3. Tie all bottles as shown in figure P1.
4. Fill bottle/jar V with 400 ml; bottle/jar IV with 350 ml; bottle/jar III with 250 ml; bottle/jar II with 200 ml; and bottle/jar I with 100 ml.
5. Tap spoon on the edges of bottles/jars in the following order: IV, II, V, I and III. Keep changing the given order to enjoy the different musical rhythms.
6. Record the difference in musical notes of all the different orders.

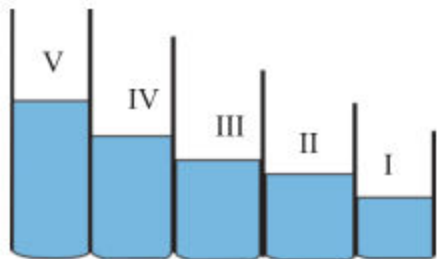


Figure P1

Activity Question:

Explain the relation between the shape of xylophone and its musical notes.

Way Forward!

Can you make a xylophone with wooden blocks and explain the relation between its shape and sound?